

CCNA 200-301 V2.0

19

First Hop Redundancy: HSRP and VRRP

Part III — IP Routing

EXAM TOPICS

3.4

LECTURER

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What you will be able to do

- **Explain** the problem a first hop redundancy protocol solves, and why routing protocols do not solve it.
- **Interpret** the operational status of HSRP and VRRP — which the blueprint asks for by name.
- **Configure** basic HSRP and VRRP with priority and preemption.
- **Diagnose** the two classic faults: both routers active, and a gateway that fails over but does not fail back.

Before we start

Chapter 15 for gateways; Chapter 10 for SVIs, which is where these protocols usually live.

Vocabulary for this chapter

first hop redundancy

virtual IP

virtual MAC

HSRP

VRRP

active

standby

master

backup

priority

preemption

group number

tracking

Each is defined where it first matters — not here.

The verb is interpret, not configure

This is what the blueprint actually asks for

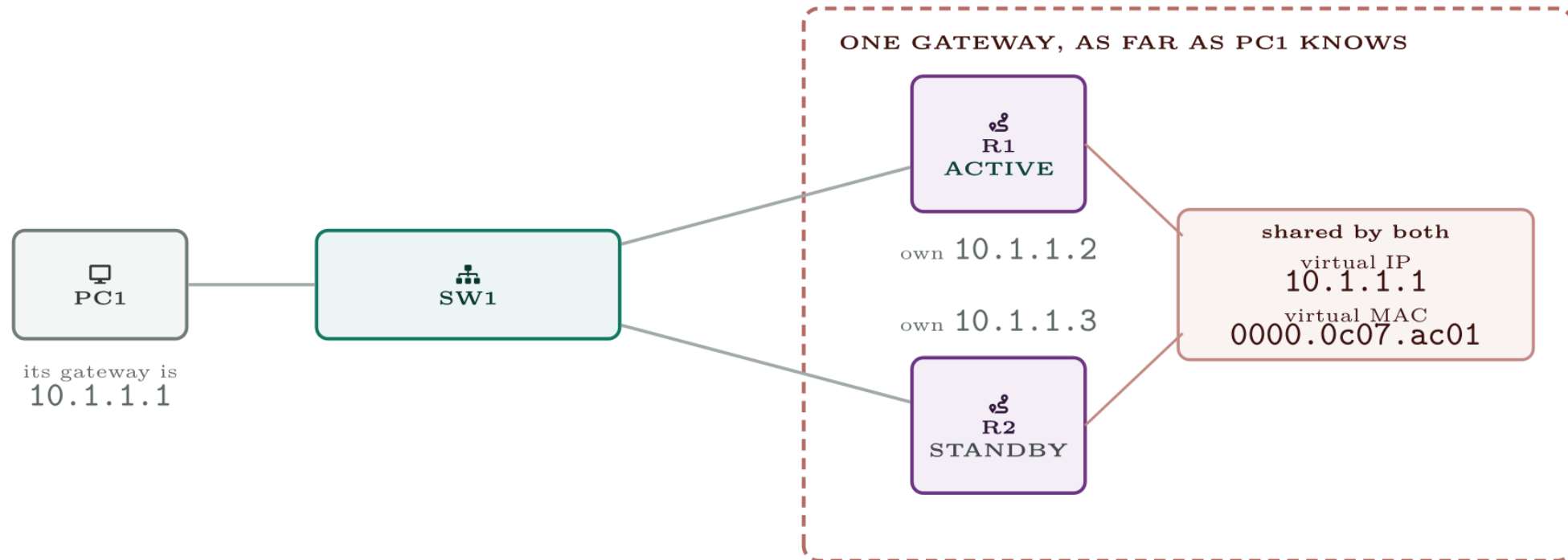
Topic **3.4** is “interpret the operational status of First Hop Redundancy Protocols (HSRP and VRRP)”. You are far more likely to be shown `show standby brief` and asked which router is forwarding, or why a failover did not happen, than to be asked to configure it from scratch. Configuration is here because you cannot interpret what you have never built — but read the verification sections hardest.

The solution: a gateway address that is not tied to a router

Two routers share a **virtual IP address** and a **virtual MAC address**. Hosts are configured with the virtual IP as their gateway and never know that two routers exist. One router is elected to answer for it; the other waits.

When the active router fails, the standby takes over *both* the virtual IP and the virtual MAC. Because the MAC does not change, the hosts' ARP caches stay valid and no host has to do anything at all. That is the whole trick, and it is why the virtual MAC matters as much as the virtual IP.

The problem



PC1 knows one address and one MAC. If R1 fails, R2 answers to *both* of them — so PC1's ARP cache is still correct and it never notices.

The host is configured once and never reconfigured. Failover moves the virtual IP and the virtual MAC, which is why nothing on the LAN has to re-learn anything.

The preemption default is the difference that catches people

HSRP does **not** preempt by default. Configure a router with priority 150, reload it, and when it comes back the other router — priority 100 — keeps the active role. Your carefully planned primary is now the standby, and everything still works, so nobody notices until the second failure.

VRRP preempts by default, so it behaves the way most people expect.

If you want a specific router to be primary, configure `standby 10 preempt` explicitly. This is the single most commonly missed HSRP command.

Hsrp on two multilayer switches, vlan 10

Configure

```
! ----- CORE1: the intended primary -----
interface Vlan10
 ip address 192.168.10.2 255.255.255.0
 ! The virtual IP -- this is what hosts use as their gateway.
 standby 10 ip 192.168.10.1
 ! Higher priority wins the election.
 standby 10 priority 150
 ! Without this, CORE1 will NOT take the role back after a reload.
 standby 10 preempt
 ! Give up the role if the uplink dies -- otherwise CORE1 stays
 ! active while having no path to anywhere.
 standby 10 track GigabitEthernet0/24 60
 standby 10 name VLAN10-GW
!
! ----- CORE2: the standby -----
interface Vlan10
 ip address 192.168.10.3 255.255.255.0
 standby 10 ip 192.168.10.1
 standby 10 priority 100
 standby 10 preempt
```

Tracking, and why it is not optional

Without tracking, the active router keeps the role as long as its own VLAN interface is up. If its *uplink* fails, it is still active, still answering for the gateway, and has nowhere to send the traffic — a black hole that the standby router cannot detect, because from its point of view the active router is perfectly healthy.

`standby 10 track Gi0/24 60` decrements the priority by 60 when that interface goes down: 150 becomes 90, which is below CORE2's 100, so CORE2 preempts and takes over. The decrement must be large enough to cross the gap between the two priorities, or nothing happens.

show standby brief

```
CORE1# show standby brief
```

```
                P indicates configured to preempt.
                |
Interface      Grp  Pri  P State      Active      Standby      Virtual IP
Vl10           10   150 P Active     local        192.168.10.3  192.168.10.1
Vl20           20   100 P Standby    192.168.10.3 local         192.168.20.1
```

show standby Vlan10

```
CORE1# show standby Vlan10
```

```
Vlan10 - Group 10  
  State is Active  
    2 state changes, last state change 04:12:19  
  Virtual IP address is 192.168.10.1  
  Active virtual MAC address is 0000.0c07.ac0a  
    Local virtual MAC address is 0000.0c07.ac0a (v1 default)  
  Hello time 3 sec, hold time 10 sec  
  Preemption enabled  
  Active router is local  
  Standby router is 192.168.10.3, priority 100 (expires in 8.048 sec)  
  Priority 150 (configured 150)  
    Track interface GigabitEthernet0/24 state Up decrement 60
```

show vrrp brief

```
CORE1# show vrrp brief
```

Interface	Grp	Pri	Time	Own	Pre	State	Master addr	Group addr
Vl30	30	110	3570	Y	Y	Master	192.168.30.2	192.168.30.1

Interpreting the status

- **Forgetting `preempt` on HSRP.** The intended primary never takes the role back.
- **Mismatched group numbers.** Two routers with different group numbers on the same VLAN both become active, each for its own group. The symptom is intermittent connectivity as hosts ARP for a virtual IP that two devices answer.
- **Mismatched virtual IPs.** Same result.
- **A track decrement too small to matter.** 150 with a decrement of 10 still beats 100. Size the decrement against the gap.
- **Configuring the virtual IP as a router's real address in HSRP.** Not allowed. VRRP permits it; HSRP does not.

Interpreting the status (continued)

- **Assuming the gateway works because the routers are up.** Test by pinging the *virtual* address from a host, then failing the active router and pinging again.

Both routers say they are active.

Users on VLAN 10 report intermittent loss. Both core switches report the active role for group 10.

Both routers say they are active.

What would you check first — and what do you expect to see?

show standby brief

```
CORE1# show standby brief
```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Vl10	10	150	P	Active	local	unknown	192.168.10.1

show standby brief

```
CORE2# show standby brief
```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Vl10	11	100	P	Active	local	unknown	192.168.10.1

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. Look at the `Grp` column: CORE1 is in group **10**, CORE2 in group **11**. They are not peers at all — they are two independent HSRP groups that happen to share a virtual IP address. Each sees no partner, so each elects itself active, and each answers ARP for `192.168.10.1` with a *different* virtual MAC (`0000.0c07.ac0a` against `0000.0c07.ac0b`). Hosts get whichever reply arrives first, and traffic splits unpredictably between two routers.

The giveaway is `Standby unknown` on both sides. A healthy pair always names its partner.

And the lesson

Fix. Correct the group number on CORE2 to 10. Within a hello interval the two will see each other, the priorities decide, and one becomes standby. Confirm that each side names the other in the **Standby** column, and that only one virtual MAC is now in use.

Fail a gateway and watch it survive

Topology. Two multilayer switches as redundant gateways for VLANs 10 and 20; an access switch; two PCs.

1. Address both cores in VLAN 10 and configure HSRP group 10 with virtual IP **192.168.10.1**, CORE1 at priority 150 with preempt.
2. Point the PCs at the virtual IP and confirm they can reach the rest of the network. Record the ARP entry a PC holds for its gateway — note the virtual MAC.
3. Run **show standby brief** on both. Identify active, standby and the preempt flag.
4. Start a continuous ping from a PC, then shut CORE1's VLAN 10 interface. Count how many pings are lost and confirm the PC's ARP entry has *not* changed.
5. Bring CORE1 back and confirm it resumes the active role. Then remove **preempt**, repeat, and observe that it does not.

Fail a gateway and watch it survive (continued)

1. Configure tracking on CORE1's uplink with a decrement of 60. Shut the uplink and confirm failover; use a decrement of 10 instead and confirm it does not.
2. Configure VLAN 20 with CORE2 as primary, so the two share the load.
3. **Extension.** Repeat the first four steps with VRRP on a third VLAN and note the difference in default preemption behaviour.

CHECKPOINT 1 of 3

Why does a host not need to update its ARP cache when the active router fails?

Discuss · then advance

Checkpoint 1 of 3

Why does a host not need to update its ARP cache when the active router fails?

Because the virtual MAC address moves with the virtual IP. The host's ARP cache still maps the gateway IP to the same MAC; the new active router simply starts answering to it. The host never learns anything happened, which is the entire point.

CHECKPOINT 2 of 3

`show standby brief` shows Standby unknown on both routers.
What does that indicate?

Discuss · then advance

Checkpoint 2 of 3

show standby brief shows Standby unknown on both routers. What does that indicate?

That the two routers are not seeing each other's hellos — so each believes it is alone. Causes: mismatched group numbers, different subnets, a VLAN or trunk fault between them, or an ACL blocking the multicast. Both then claim active, and two devices answer for the same IP.

CHECKPOINT 3 of 3

A router is configured with priority 150 and a tracked interface with decrement 10. Its peer is at 100. The tracked interface fails. What happens, and is it what was intended?

Discuss · then advance

Checkpoint 3 of 3

A router is configured with priority 150 and a tracked interface with decrement 10. Its peer is at 100. The tracked interface fails. What happens, and is it what was intended?

The priority drops to 140, which is still above the peer's 100, so **nothing happens** — the router keeps the active role despite having lost the tracked interface. That is almost certainly not what was intended: the decrement must be large enough to take the priority *below* the peer's, so 51 or more.

What to take away

- Hosts have one static gateway address, so redundancy has to be provided below them: two routers share a virtual IP *and* a virtual MAC.
- HSRP is Cisco, uses active and standby, and **does not preempt by default**. VRRP is standard, uses master and backup, and does.
- Priority decides; preemption decides whether the better router takes the role back.
- Track the uplink, with a decrement large enough to cross the priority gap, or the active router can become a black hole.
- Interpret `show standby brief` by columns: priority, the **P** flag, state, and who is named as partner. `Standby unknown` means the two are not peering.

Where we got to

- Hosts have one static gateway address, so redundancy has to be provided below them: two routers share a virtual IP *and* a virtual MAC.
- HSRP is Cisco, uses active and standby, and **does not preempt by default**. VRRP is standard, uses master and backup, and does.
- Priority decides; preemption decides whether the better router takes the role back.
- Track the uplink, with a decrement large enough to cross the priority gap, or the active router can become a black hole.

NEXT

Chapter 20 — DHCP: Client, Server and Relay